

"This study may lead to future computers that can operate within the human body, interacting with its biochemical environment to yield far-reaching biological and pharmaceutical applications."

Not to downplay the stupendous achievement, we should emphasise that such "computers" are not very different from the one we discussed—Adleman's DNA computer solving the TSP. The future implications for DNA computing that such news items predict are typically vague and broad. They say nothing about what DNA computing will be like, and whether you'll see a DNA desktop anytime. But that's what we really want to know—enough of the test tubes!

Self-Assembly

According to a Web definition, self-assembly is "the fundamental principle that generates structural organisation on all scales—from molecules to galaxies. It is defined as a reversible process in which pre-existing parts or disordered components of a pre-existing system form structures of patterns." According to another definition, self-assembly is "a branch of nanotechnology in which objects, devices, and systems form structures without external prodding."

Adleman's work was just the beginning. His 1994 paper basically explained how one-dimensional self-assembly of DNA—remember how the molecules arranged themselves physically according to the inputs?—could work like a FA. That was the foundation for connecting computation and self-assembly of DNA.

His work inspired Erik Winfree, in 1996, to propose that *two-dimensional* self-assembly of DNA can perform universal computation, with a set of 'molecular Wang tiles' as the 'program'. Winfree's proposal used Hao Wang's idea that computation had something to do with tiling. If Winfree's proposal was right, DNA would be a material that could, well, be used in a DNA computer.

We need to get our hands dirty again. What is a "universal computation"? What are molecular Wang tiles? How did Wang relate computation to tiling? What is tiling, anyway? There are straight answers to all these.

Winfree And Wang

A universal computation means any computation at all—just as the phrase implies. The full form is "Turing-universal," which refers to Turing machines. A Turing machine is the underlying conceptual 'machine' behind all computers today—whatever a Turing machine is capable of, any computing machine is capable of.

To put it simply,

A Set Of 13 Wang Tiles—Do They Tile The Plane?



Wang tiles are equal-sized squares with a colour on each edge. To tile the plane using a set of Wang tiles, the colours on the edge must match. This is a set of 13 Wang tiles that can tile the plane aperiodically—meaning the tiling pattern doesn't repeat

think of a Turing machine as something that can do everything a general-purpose computer can do.

A tiling is an arrangement of a few basic shapes, called tiles, that fit perfectly together and stretch any area you can imagine. Think of bathroom tiles—they're all square, and they fit perfectly together for any extent you can imagine. Bathroom tiles, however, aren't very interesting, because they're all square! Think of, say, a rectangle and two different kinds of triangles. If you had an infinite supply of these, would you be able to tile any area? Of course, that depends on the shape of the triangles; if you can, that set (of the rectangle and the triangles) is said to "tile the plane," if not, then the set does not tile the plane. And if the tiling pattern can repeat, it's called a periodic tiling.

Given a set of tiles, it was expected by scientists that one should be able to say whether they can periodically tile the plane. When Wang investigated the problem—called the "tiling problem"—in the 1960s, he discovered it was unsolvable. That is, given the set of polygons, it could be *proved* that there was no way of telling whether the set would tile the plane or not.

The tiling problem thus 'reduced' to the Halting Problem of Turing Machines. That's the computer-science way of saying that if you could solve the tiling problem, you could achieve the impossible. However, it also means that a machine (or whatever) that could go about trying to tile the plane is theoretically as powerful as a regular computer. For example, your computer, given a set of tiles that *does* tile the plane, will be able to find that it tiles the plane. In other words, dealing with tiles was proved to be the equivalent of computing in general!

Now, there is a close relationship between the way the plane is tiled and the way *crystals* grow. (Crystals start forming when a seed is put into a gel, for example: the gel starts crystallising from the seed outwards. The seed can even be a speck of dust.) Winfree asked if crystal growth had the potential to compute as powerfully as tiling did. To answer *that* question, one needed the ability to design *molecular* Wang tiles. (Wang tiles are square tiles coloured on the edge, which pose the question whether the plane can be tiled using them.)